(a) HiLoStem: Pre-Token Detail Injection

Inject high-resolution edge/texture cues before tokenization

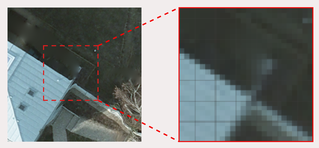
Input patch

$X \in \mathrm{R}^{3\times H\times W}$



Why HiLoStem?

Vanilla Patch Embedding



Details weakened
Boundaries attenuated

HiLoStem



Details preserved
Edges retained

High-resolution branch (stride-1)

Conv
3×3, s=1
+ GELU

DWConv
7×7
(group=C)
+ SiLU

SE Gate
$s = \sigma(W_2\,\delta(W_1\,GAP(f)))$
$y = f \odot s$

Low-resolution branch (stride-4)

Patch Embedding
Conv 4×4, s=4

$\boldsymbol{P}$
$\boldsymbol{P} \in \mathrm{R}^{C\times H/4\times W/4}$

AvgPool
4×4, s=4

$\tilde{\boldsymbol{y}}$
$\tilde{\boldsymbol{y}} \in \mathrm{R}^{C\times H/4\times W/4}$

Residual Fusion
$E = P + \gamma\tilde{y}$
(learnable γ)

$\boldsymbol{E}$
$\boldsymbol{E} \in \mathrm{R}^{C\times H/4\times W/4}$

Flatten & LayerNorm

Tokens
$(N = H/4 \times W/4,\ C)$

Feature map · High-res branch · Low-res branch

(b) AxialLite: Decoder Context Propagation

Propagate information along horizontal and vertical axes for cross-region interaction at moderate resolution

Input from decoder (Stage-3 feature)

$X_3 \in \mathrm{R}^{C\times H\times W}$

1×1 Conv (In)
Project $C \rightarrow 2C$

$Z_0 \in \mathrm{R}^{2C\times H\times W}$

Parallel axial depthwise branches

Horizontal axial branch

DWConv 1×k (k=7)
(groups = $2C$) + SiLU

Vertical axial branch

DWConv k×1 (k=7)
(groups = $2C$) + SiLU

Z_x Z_y

$Z = Z_x + Z_y$

1×1 Conv (Out)
Project $2C \rightarrow C$

$Y = X_3 + Conv_{1\times1}(Z)$

Output to decoder (Stage-3 output)

$Y \in \mathrm{R}^{C\times H\times W}$

Feature map · Horizontal branch · Vertical branch · Element-wise add